

Tutorial_7_Distributed_Development

Tutorial 7: Distributed Development with Workers

Duration: 30 minutes **Level:** Advanced

Overview

Scale development with distributed SSH workers: - Add remote workers - Auto-installation - Task distribution - Health monitoring

Step 1: Prepare Worker Machines

```
# On each worker machine
# Ensure SSH access and Go installed
```

```
# From control machine
ssh worker1.example.com
ssh worker2.example.com
```

Step 2: Add Workers to HelixCode

```
# Add first worker
helixcode worker add \
  --host worker1.example.com \
  --user helix \
  --key ~/.ssh/id_rsa

# HelixCode automatically:
# 1. Connects via SSH
# 2. Detects hardware (CPU, RAM, GPU)
# 3. Installs HelixCode CLI
# 4. Starts worker agent

# Output:
# ✓ Connected to worker1.example.com
# ✓ Hardware detected: 16 CPU, 64GB RAM, NVIDIA RTX 4090
# ✓ Installing HelixCode CLI...
# ✓ Worker registered: worker-001

# Add more workers
```

```
helixcode worker add --host worker2.example.com --user helix --  
key ~/.ssh/id_rsa  
helixcode worker add --host worker3.example.com --user helix --  
key ~/.ssh/id_rsa
```

Step 3: View Worker Pool

```
helixcode worker list
```

#	ID	Host	Status	CPU	RAM	
#	worker-001	worker1.example.com	healthy	16	64GB	RTX
	GPU 4090	Tasks 2/10				
#	worker-002	worker2.example.com	healthy	32	128GB	RTX 4090
	x2	5/20				
#	worker-003	worker3.example.com	healthy	8	32GB	
	None	1/5				

Step 4: Distribute Tasks

```
# Create parallel tasks  
helixcode task create "Build Linux binary" --worker worker-001  
helixcode task create "Build macOS binary" --worker worker-002  
helixcode task create "Build Windows binary" --worker worker-003
```

```
# Or let HelixCode auto-assign  
helixcode task create "Run test suite" --auto-assign
```

```
# HelixCode considers:  
# - Worker capabilities (OS, arch, GPU)  
# - Current load  
# - Task requirements
```

Step 5: Monitor Execution

```
# Watch all workers  
helixcode worker monitor
```

```
# Live dashboard:  
# ┌ worker-001 ────────────  
# │ CPU: ██████████░░░░ 80%  
# │ RAM: ██████████░░░░ 60%  
# │ Task: Building Linux binary...  
# │ Progress: 75%  
# └───────────────────┘
```

Step 6: Collect Results

```
# Tasks complete automatically
# Results synced back to control machine
```

```
helixcode task results task-001
```

```
# Output:
# Task: Build Linux binary
# Status: Completed
# Duration: 2m 34s
# Artifacts:
#   - bin/app-linux-amd64
#   - bin/app-linux-arm64
```

Step 7: Complex Workflow

```
# Distributed test execution
helixcode test --distributed \
  --workers worker-001,worker-002,worker-003
```

```
# HelixCode:
# 1. Splits test suite
# 2. Distributes to workers
# 3. Runs in parallel
# 4. Aggregates results
```

```
# Time: 45 minutes → 15 minutes (3x speedup)
```

Step 8: GPU Acceleration

```
# Use GPU worker for AI tasks
helixcode generate --worker worker-002 \
  --model ollama/llama3:70b \
  --prompt "Generate complete user authentication system"
```

```
# Worker-002 uses local GPU for inference
# No API costs!
```

Configuration

```
workers:
  ssh:
    timeout: 30s
    max_retries: 3

  auto_install: true
```

```
health_check_interval: 30s

# Task assignment strategy
assignment:
  strategy: "capability" # capability, load, round_robin
  prefer_gpu: true
  prefer_local: false
```

Results

- **Parallelization:** 3x+ faster builds and tests
 - **Scalability:** Add workers as needed
 - **Cost Efficiency:** Use existing infrastructure
-

Continue to [Tutorial 8: Using Plan Mode](#)

Sources verified

Sources verified 2026-05-29: <https://ollama.com/library> (Ollama model library) ; <https://go.dev/dl/> (go1.26.3 latest stable Go) ; project go.mod (inner go 1.26) + CLAUDE.md §3.1. **NEGATIVE FINDING** — the example tag `ollama/llama3:70b` references Meta Llama 3 70B, which is still present in the Ollama library but is no longer the latest; newer tags `llama3.1` (8B/70B/405B), `llama3.2` (1B/3B), and `llama3.3` (70B, ~Llama-3.1-405B-class quality) are now available. Per CONST-036 model IDs are LLMsVerifier-sourced at runtime — confirm available tags with `ollama list/helixcode llm models list` rather than relying on the example string. Left as-is (illustrative, still valid) but flagged.